

GROUP #3: Honkshoe

Element E - Application of STEM - E1: Application of STEM Concepts to Design

In our project we are addressing the lack of ergonomics at Ms.A.V workspace, where we applied various STEM concepts to ensure the design solution would be effective, functional, and feasible. Down below is a breakdown of the concepts used:

- 1. Engineering:** With our design we will be using engineering techniques that will ensure and allow the desk to be stable and will be able to withhold various weights without breaking. We will be also using engineering principles to choose the best materials that maximizes the structural integrity of our group's prototype, but also still maintains the factor of affordability. We will use a process such as material selection, to ensure that when choosing the type of wood for our design or other materials are being chosen due to that they meet the requirement of durability, having a lot of strength to withhold various weight, affordable, and maximizing the structural integrity of the desk. Also, after much research we will learn about using finite element analysis (FEA), and how to apply it so that when building we can identify weak points in the structure and be able to fix them. Also, we will need to understand beam deflection formulas to make sure that our desk surface does not sag or bend from heavy weights. Finally, we will apply and learn more about strength-to-weight ratio to apply it to the material selection when it comes to wood, metal, other materials, to ensure an efficient and effective product.
- 2. Physics:** We have to be precise with the creation of our desk for Ms. A.V to make sure it is balanced and does not have any possible issues such as easily tipping over and possibly getting damaged. We will have imbalances in weight throughout the desk, so physics and basic calculus is to be used to properly plan what is needed out and to ensure a safe and functional product. These calculations and physics topics that will be applied are Newton's Laws of Motion to ensure the stability and evenly weight distribution throughout the entire desk and we would have to understand what stress-strain relationship is and learn how to apply it to the building of our prototype to ensure that materials can withstand force without deformation. We also will need to make sure to properly apply the physics of resin pouring and use it to make sure there are no air bubbles or dips in the final product and result. Trying to correct anything afterwards would be very difficult and time consuming.

- 3. Electrical Engineering:** In the final design, we hope to add an electrical outlet extension, meaning we have to be careful with the electrical risks it may possess and prevent any misdoing. During building and incorporating this outlet to our design we must ensure that there is proper grounding and insulation and learn techniques to ensure that this is added to our design that we ensure functionality of the outlet and it is safe for our stakeholder to use. We also use online programs to properly design and plan out a prototype while taking measurements and elements in consideration to provide us and our stakeholders with a vision of the final product. Modern and traditional technology is to be used for the creation itself and communication of it.
- 4. Mathematics:** In order to effectively and efficiently plan out our design, we will have to conduct exact measurements and predict overall costs and needs of the design as a whole. The desk itself has varying measurements, and the drawers included with it need to have measurements specifically articulated for it to be a precise fit. We will apply trigonometry to correctly calculator angles for bracing, apply cost analysis to determine the cost of our materials, and overall learn and apply height-to-width ratio to ensure that our desk has correct proportions in its design. Anything that is inaccurately measured or predicted may lead to the failure of the design, such as not being an even creation, or not having enough resin or wood to complete the design. We have to keep into consideration the space in Ms. A.V 's available area to make sure our design and prototype does not exceed our limitations and fits her wants.
- 5. Chemistry (Epoxy Resin Process):** When it comes to incorporating the resin art feature to our standard computer desk for our stakeholder, chemistry will be involved where we will be using epoxy resin which is a type of resin for artistic and protective finishes on builds and designs. When it comes to using epoxy resin we will have to apply and understand the polymerization reaction, and overall be cautious of factors like temperature, humidity, and air, which can affect the overall chemical mixture where they can be bubbles or overall too weak. Furthermore, a proper chemical mixing will ensure that Ms.A.V desk will have smooth, glossy finishes that enhance the aesthetics factor of the desk and ensure durability and its longevity. This chemical based approach allows for the desk to be efficient and visually appealing where it can last for a long use of time

6. **Psychology (Aesthetics):** While an aesthetically pleasing desk might seem contradictory to the organized working environment that our stakeholder requested, it's actually the opposite. According to The Guardian in a 2016 report, individuals who had the opportunity to have art and plants in their work area had worked nearly 15% quicker than office workers with no art displayed ([link](https://www.theguardian.com/careers/2016/jan/21/art-works-how-art-in-the-office-boosts-staff-productivity)). This percentage doubled when these individuals were able to control the arrangement and selection of these art pieces. Art in working environments helps motivate workers, surrounding them in an environment that encourages them rather than a sparsely decorated environment that can feel dull and lifeless. The team found that people who worked in the enriched office worked about 15% quicker than those in the lean office and had fewer health complaints – this figure then doubled for people who worked in the empowered space. <https://www.theguardian.com/careers/2016/jan/21/art-works-how-art-in-the-office-boosts-staff-productivity><https://hbr.org/2022/01/a-beautiful-workspace-can-make-your-job-more-joyful>

Element E - Application of STEM - E2

1. **Cost & Feasibility:** All of the materials from this project will either be donated by team members or mentors, or funded by the entire team. This means that we have to be conscious of the cost of the materials that we need and any alternative resources we could utilize to build this desk.
2. **Practicality:** After interviewing our stakeholder, Ms.A.V and getting her feedback we understood what she wants for her desk and we determined that we must make a computer desk that needs to be functional, durable, and aesthetically pleasing. This resulted in us applying engineering principles and creating a 3D design because it would simulate what a real computer desk looks like from all angles allowing for a more immersive experience and allowing both us and the stakeholder to go more in detail when creating the desk and making sure that every aspect of it satisfies the stakeholder, Ms.A.V . Furthermore, our computer desk with a second level for our monitors will allow for organization, aesthetics, and functionality for our stakeholders where they will be able to focus and be more productive where their ergonomics will improve because of this desk.

3. **Critical-Thinking:** Using our current knowledge and information we have gained we could build a desk that can work best for Ms. A.V . We of course ask a lot of questions for our stakeholders as we write down the details to then make decisions about how the deal would look like. Using new or old information we can analyze and make desk designs that would not only be aesthetically pleasing, but also beneficial to our stakeholder. With the design we make after discussions with each other and our stakeholder and mentors we can make decisions that would affect the aesthetic and size when building our desk.
4. **Interactive & Collaborative:** The success of this project relies heavily on our interactions with our mentors and stakeholders. Since we are building a desk specifically for Ms. A.V , we have to communicate consistently with her at all the different stages of the project. By keeping her informed about the process of creating her desk, we can ensure that she is informed about all the different decisions we make on the desk and that she knows about any issues we have when adhering to her wants. We also have to communicate thoroughly with our mentors during this project. Since we have no experience with building a piece of furniture like this, it's important that we have reliable mentors that are able to guide us through the process. Our mentors teach us about the different aspects that are involved with furniture making and they can also help provide materials for our project. Lastly, we have to communicate with our other team members. Since we are all working together to complete a single goal, it's important that we communicate with each other.
5. **Problem-Solving:** In our community projects that involve fixing up an old desk we plan to create a desk that is best suited for Ms. A.V . She has many problems with her current desk that includes limited space, uncomfortable, old, and unappealing. Using problem solving skills we can make a more customizable desk for Ms. A.V that would satisfy her needs and wants. It is important to use this skill in order to create a more suitable area for Ms. A.V to work. Not only could we create a desk for Ms. A.V , but we could create a desk that could be used in many areas.
- 6.

Element E - Application of STEM - E3

We consulted with our two mentors Mr. R.N and Olivia's father, who both have experience in carpentry and woodworking. Mr. R.N is our sourcer for our black walnut oak wood and confirmed that the measurements provided would be more than sufficient and possible

with a few tweaks. His advice for the inner measurements and material of the non-visible wood were things both him and Olivia's father suggested and agreed with, having the idea of using cheaper plywood for the walls of the inner pull out drawers with examples to see how it looked.

The resin idea and material were all approved by Mr. R.N, who would be helping up pour and make sure of the integrity of our final product. Olivia's father does not have much experience in that area, so left the judgement to Mr. R.N himself. He acknowledged that the resin would be a nice and realistic touch to our prototype, making it stand out while also mentioning different ways to pour and design the resin.

Overall, both of our mentors are in agreement with our plans and design, both deeming them acceptable and sustainable. They know that they have to overlook our whole operation, and know that what we plan on doing are within their constraints as well.

Element E - Expert Credentials and Selection Application of STEM - E4

Mr. T.B, Olivia's father, is a mechanical technician who has extensive experience with woodworking. His understanding on how to create furniture, knowledge on construction, and the fact that he's easily contactable make him a valuable mentor in this project. He is able to assist us throughout the whole project as we are working in his work area to create our desk. Mr. T.B can not only provide us materials and tools, but give us detailed feedback, suggestions, and alternatives in real time.

Mr. R.N has been a teacher at Eleanor Roosevelt for a number of years. Currently, he runs the school's Construction club, and teaches an engineering class and also an architectural drafting class. Mr. R.Ns extensive knowledge on construction and experience with working with resin make him an ideal mentor for this project. He's able to give us feedback as we're constructing our desk and also fills in the gap in an area where Mr. T.B isn't experienced with working with resin.

Element E - Application of STEM - E5

Do you see any of the aspects of the desk being difficult to recreate in real life?

Olivia's father first questioned the size of the main desk which is Width: 50 in, Depth: 20 in, Height: 30 in. He said that building a desk this size will be difficult because it might become heavy due when the resin is added. He said that the overall design is doable. He told us to plan

the building and where to put every part because it is easy to make many mistakes that might not be able to be fixed.

Ms. R.N says that while there's nothing too unrealistic about our model and sketches, we might struggle a bit as a group with the actual process of building a desk. Since our desk is a functional desk that also serves an aesthetic purpose, he wants to make sure that we have planned carefully and thoroughly about our project. He mentioned a couple different methods on how we could connect the desk such as joinery and pocket hole screws.m

Is there anything you would change about the model?

Olivia's father said that he would make the open cabinets into pull out drawers. He said that the idea is nice, but for someone who works at their desk most of the time for work there should be cabinets for our stakeholder to store files or other work related means. Besides two open cabinets we should make three cabinets. He said since the size is 30 inches, we could do 2 cabinets, but 3 cabinets would be better because it allows for more storage. Olivia's father said that the model looks really nice, but he said that he would change the placement of the charging outlet. The places he suggests that we could move are underneath the desk, underneath the monitor tray(if we chose to increase the height), or on the wall.

Mr. R.N suggests that we should change the legs of the table. He suggests cutting the legs 'waterfall style'. As he explains it, you take one piece and cut it at a 45 degree angle so that the table leg can fold up on itself at a 90 degree angle. He says that this will add more structural integrity for our desk and adds more floor connection to the desk.

He also says that we could add a connector to the legs for stability.

Do you have any general feedback?

Olivia's fathers feedback on our model is that it looks great. A question that he has is if we were to make the cabinets out of resin too. He believes that it would look better without resin cabinets, but that is up to Ms. Vaquez. He also said that if we were to do resin cabinets, the cost would be higher.

Mr. R.N's was impressed with our model. He said that as an engineering teacher and someone interested in wood furniture, he sees that the project is a spearheading experience that can

help develop construction clubs but also transform engineering classes as the projects become more sophisticated, practical, and potentially entrepreneurial.

Are there any materials that can be replaced on the model with an alternative option? (Such as outlets, shelves, etc.)

Olivia's father said that he doesn't have any suggestions about items being replaced on the model. He said that most of our items are needs and wants of our stakeholder so it is necessary for them to be added. He also said that since most of this stuff is being donated like the wood by Mr. R.N should not be hard to get.

Mr. R.N recommends a power strip as a potential alternative to the outlet. He says that a power strip could blend in with the aesthetics of our desk and can also be placed under the monitor shelf. His initial idea on the outlet was that the cords of the outlet would be fine overall as long as it is tucked away but he would like to take a closer look at it so he's able to give more accurate thoughts and feedback.